

Semigroup / Compositional

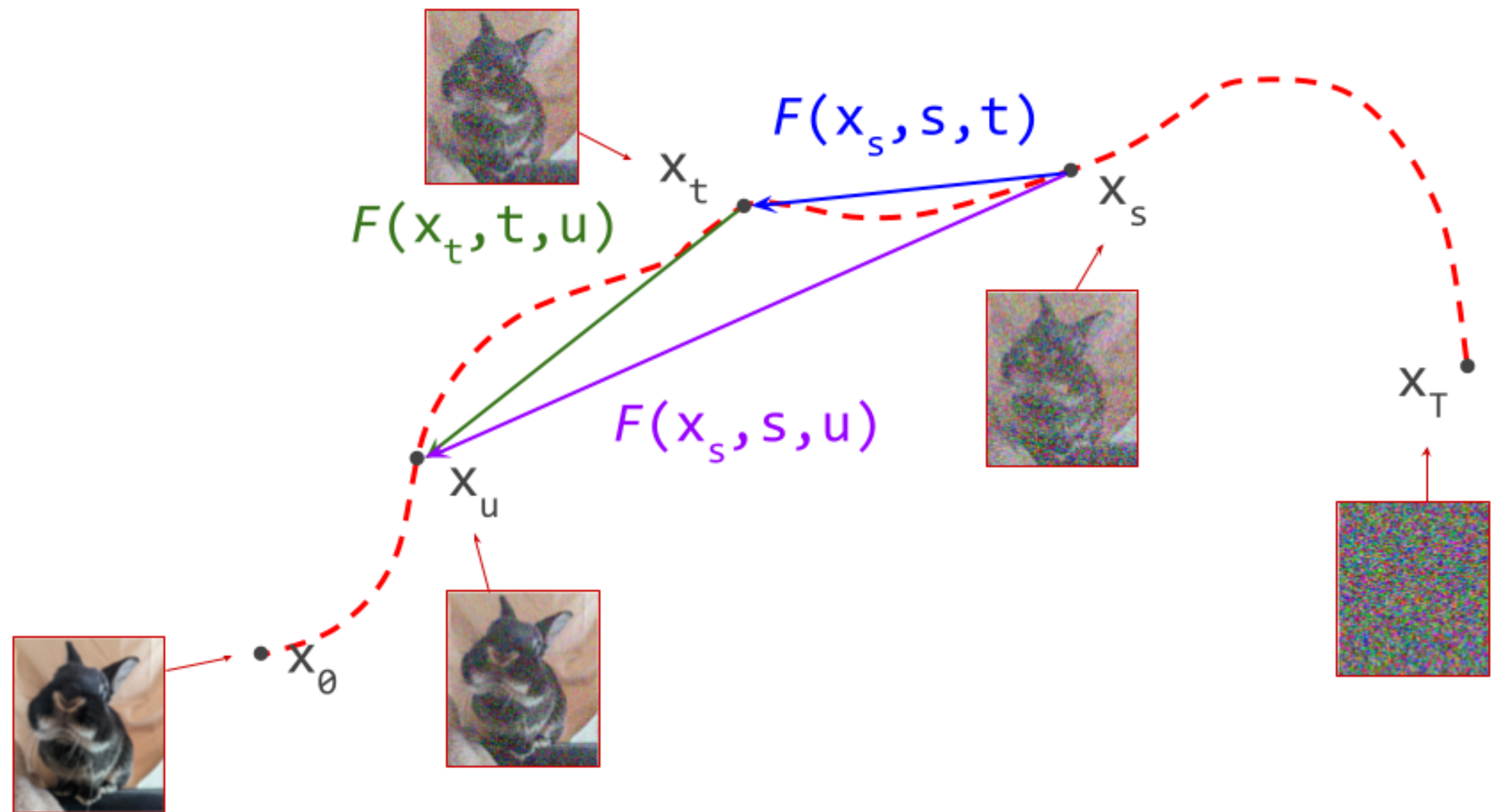
Flow Map: $F(\mathbf{x}_s, s, t) \approx \mathbf{x}_s + \int_s^t \mathbf{v}(\mathbf{x}, u) du$

Semigroup / Compositional property

$$F(\mathbf{x}_s, s, u) = F(F(\mathbf{x}_s, s, t), t, u) = \mathbf{x}_u$$

One step = two intermediate steps

Examples: *Shortcut Models*



Eulerian perspective

Flow Map: $F(\mathbf{x}_s, s, t) \approx \mathbf{x}_s + \int_s^t \mathbf{v}(\mathbf{x}, u) du$

Eulerian consistency

$$\frac{\partial}{\partial s} F(\mathbf{x}_s, s, t) = \frac{\partial}{\partial s} F(\mathbf{x}_s, s, t) + \nabla_{\mathbf{x}_s} F(\mathbf{x}_s, s, t) \mathbf{v}(\mathbf{x}_s, s) = 0$$

Match the end point $\mathbf{x}_t$ starting from different $\mathbf{x}_s$

Examples: Consistency Models, MeanFlow

